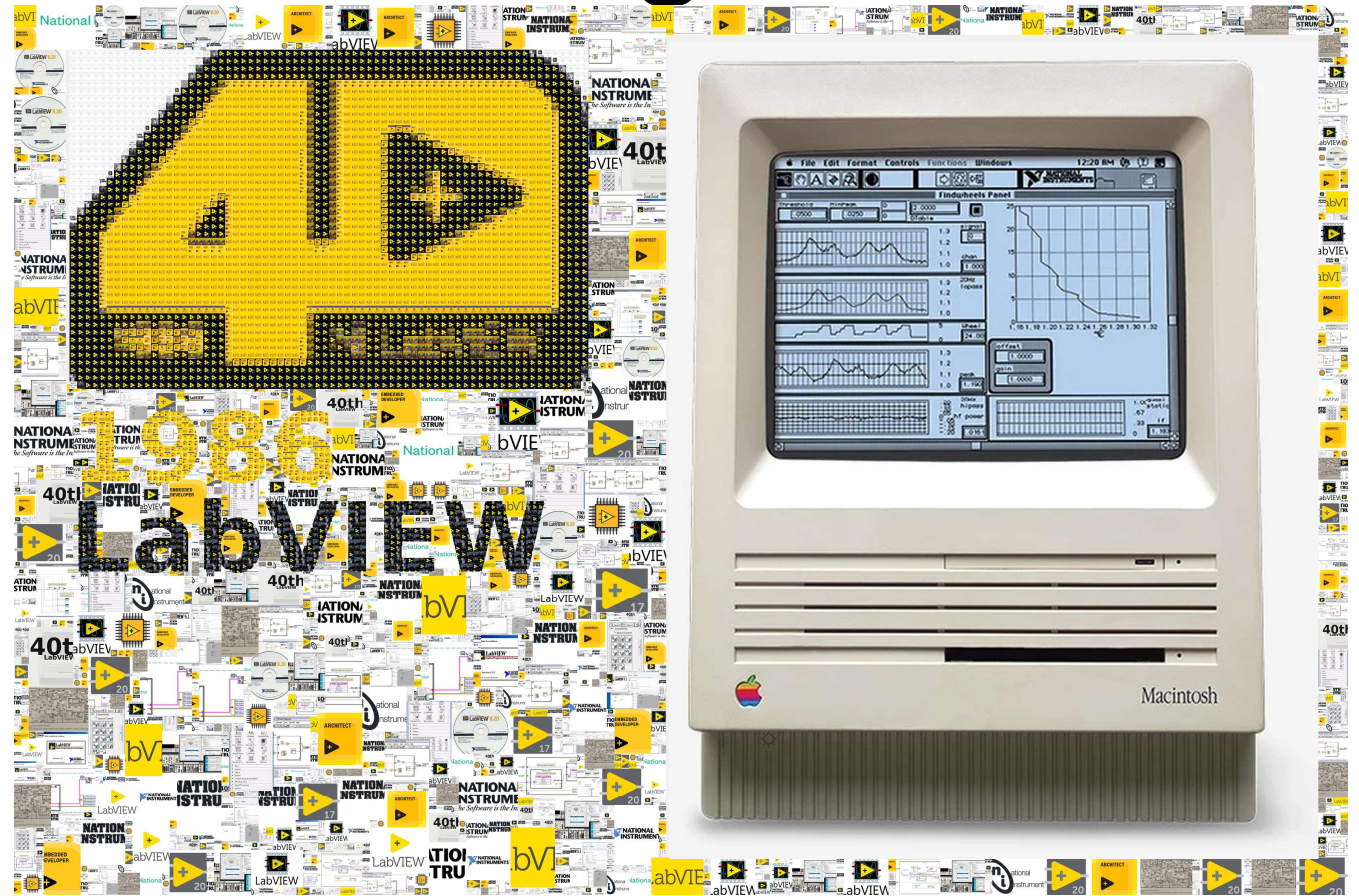


LabVIEW and AI Together



Zach Olson

BDM & LabVIEW Champion | Cyth Systems



Evolving system design and validation

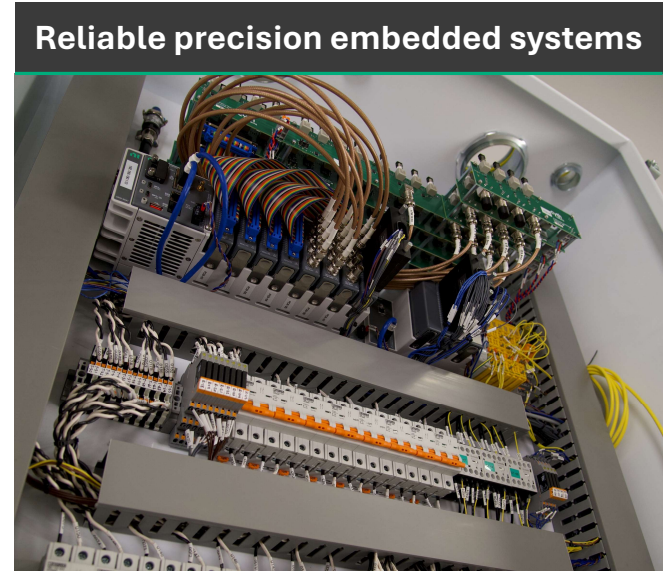
Use more AI tools



- Speed of code generation
- Leveraged industry knowledge
- Do more with less



Reliable precision embedded systems



- Determinism and reliability
- Compilation & compute toolchains
- Validated, benchmarked, compliant systems

How can we get the best of both worlds?

Agenda

Application space

Why not just use Vibe Coding – Where LabVIEW and LabVIEW RT/FPGA excel

Technical foundation

Platform, toolchain, ML vs. model-based inference

Implementation Tutorial

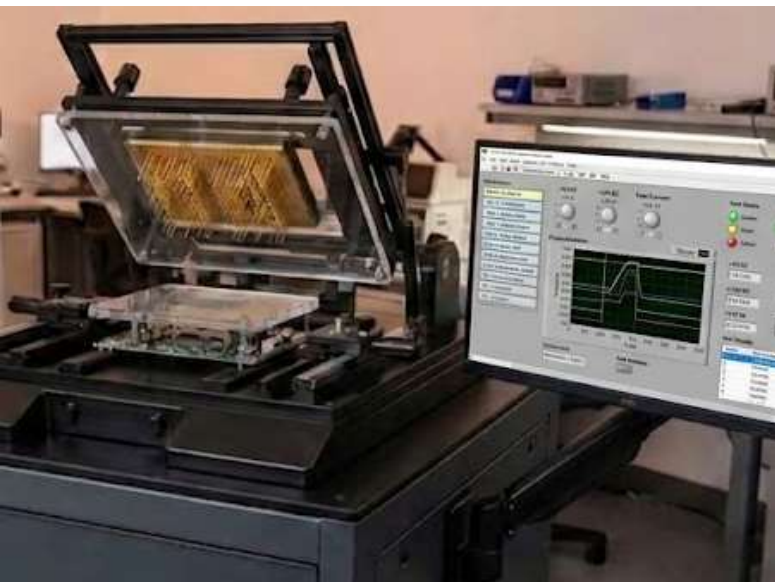
Walk through a project

System Validation

Simulation, test bench setup, benchmarking AI-generated code

Advanced topics

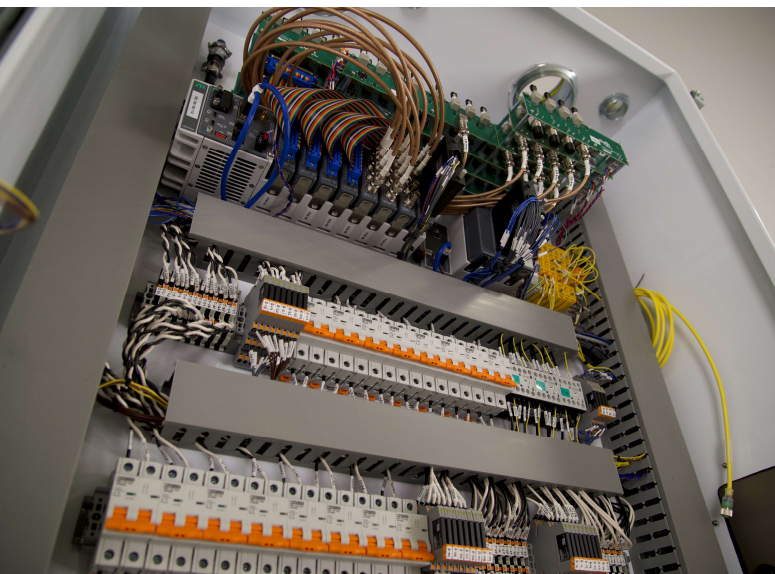
Where this is going and next steps



Cyth Systems Overview

Company overview

- Headquartered in San Diego, CA with U.S.-wide coverage
- 25+ years delivering automation solutions
- Full spectrum NI Partner



Application Expertise

- Test & measurement automation
- Precision automation & monitoring
- Embedded design & engineering
- LabVIEW & TestStand consulting services



Application Space

Where the NI platform excels

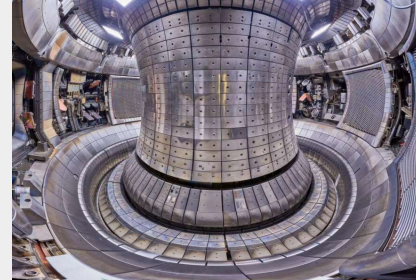
Precision Control & Monitoring



BioTech & Lab Equipment



Advanced Medical Devices



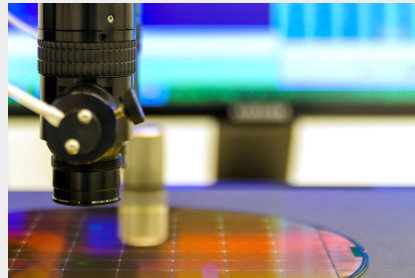
Power Gen System



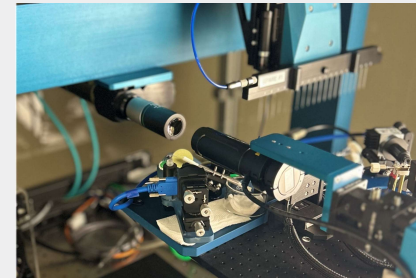
Modern Grid & DER



O&G Equipment



Materials Research & Fab



Precision Manufacturing

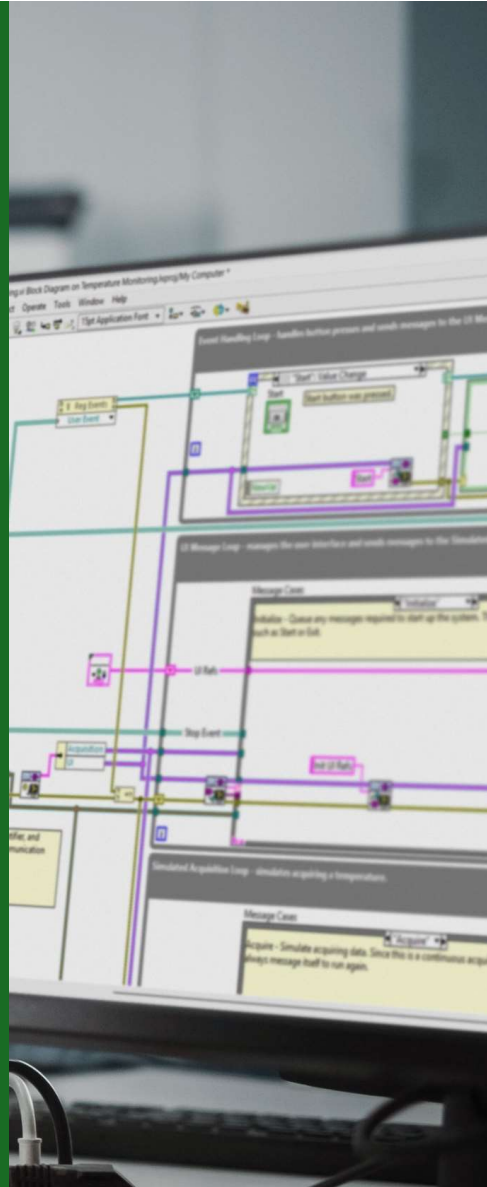


Autonomous Machines

Technical Foundation

Hardware platform, software tools, AI
code gen practices

DEMO



LabVIEW Nigel Demo

Hardware aware



AI Code Generation Tools

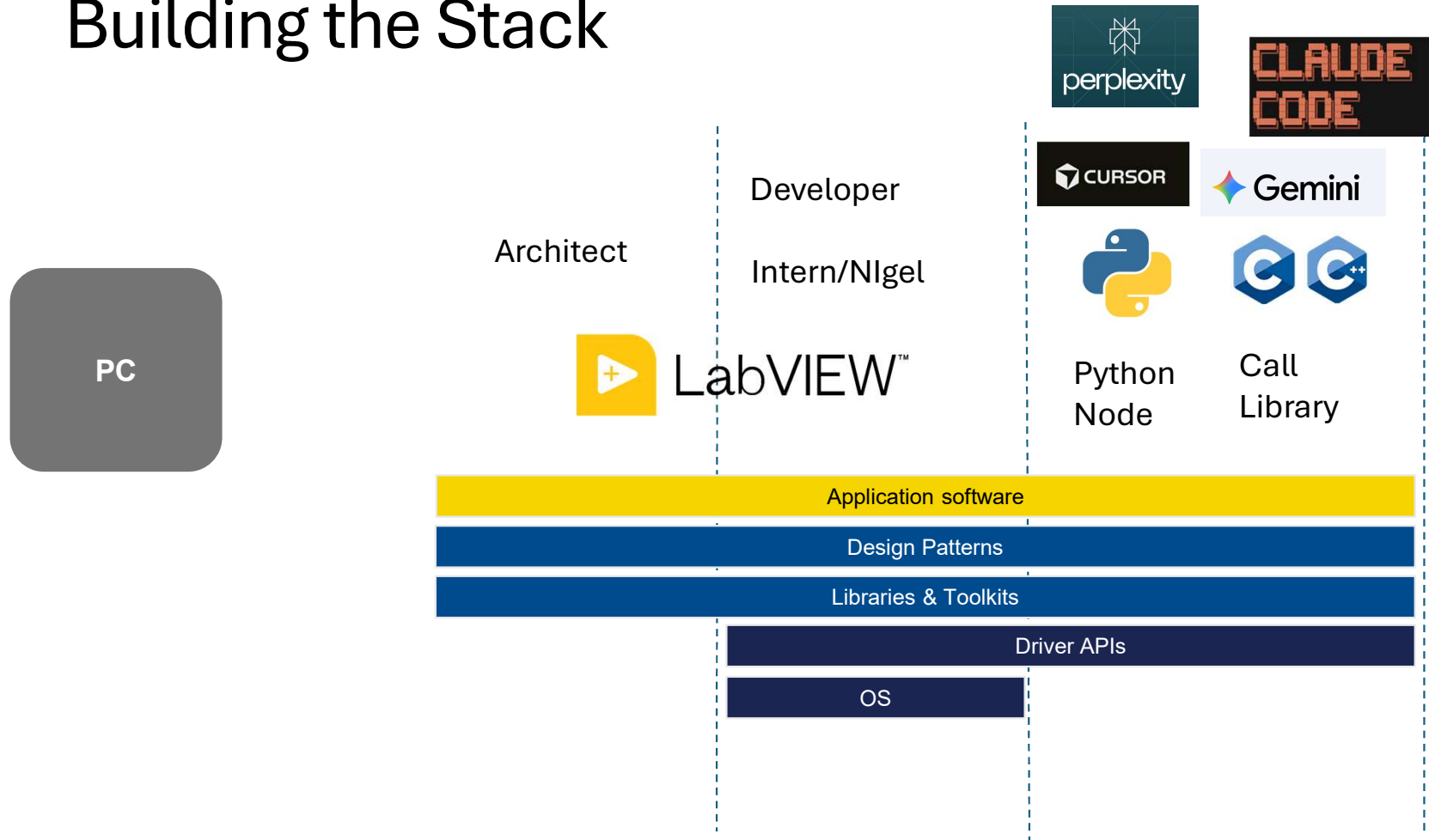


1. Learning how to prompt and build context
2. Source code control & integration
3. Test and test often

Implementation Tutorial

Let's walk through a project

Building the Stack



Platform I/O

Platform Hardware – Measure, Generate, Interface



Measure

- +/-0.25VDC up to 480VAC
- Digital (TTL, 24VDC, sinking/sourcing)
- Counters
- Current
- Temperature (TCs, RTDs)
- Load, pressure, torque sensors
- Strain gages (bridge varieties)
- Accelerometers (IEPE)
- Microphones (IEPE)
- Galvanically isolated meas.
- String pots, line pots
- Proximity probes
- pH probes
- Custom

Generate

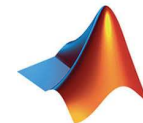
- +/-10V output
- Current (0-20mA)
- Digital (TTL, 24VDC)
- Relay module
- Generate PWM
- Simulate sensors/signals

Interface

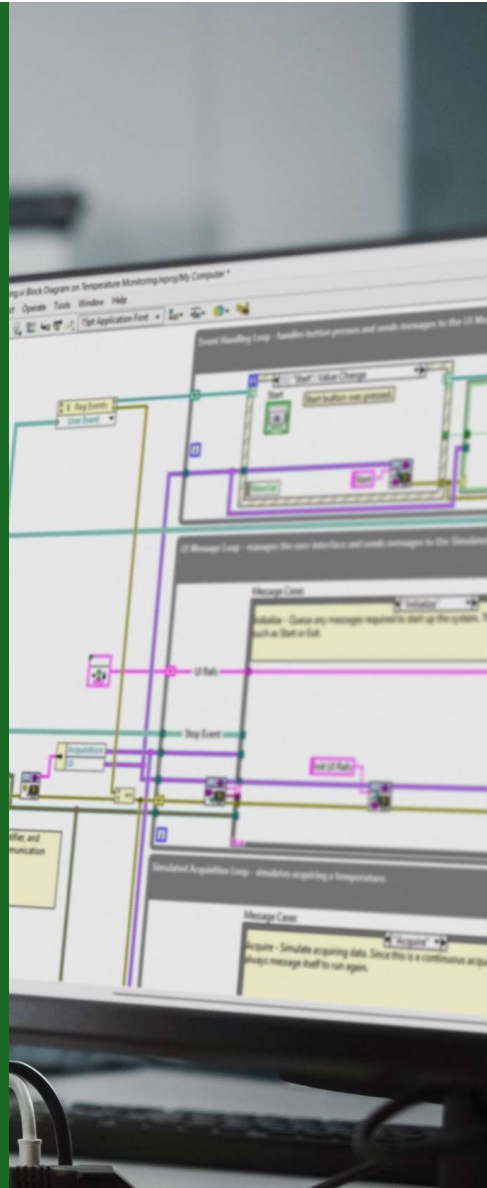
- RS232 / 485
- I2C/SPI
- Ethernet
- EtherCAT
- Modbus
- Custom



LabVIEW™



DEMO

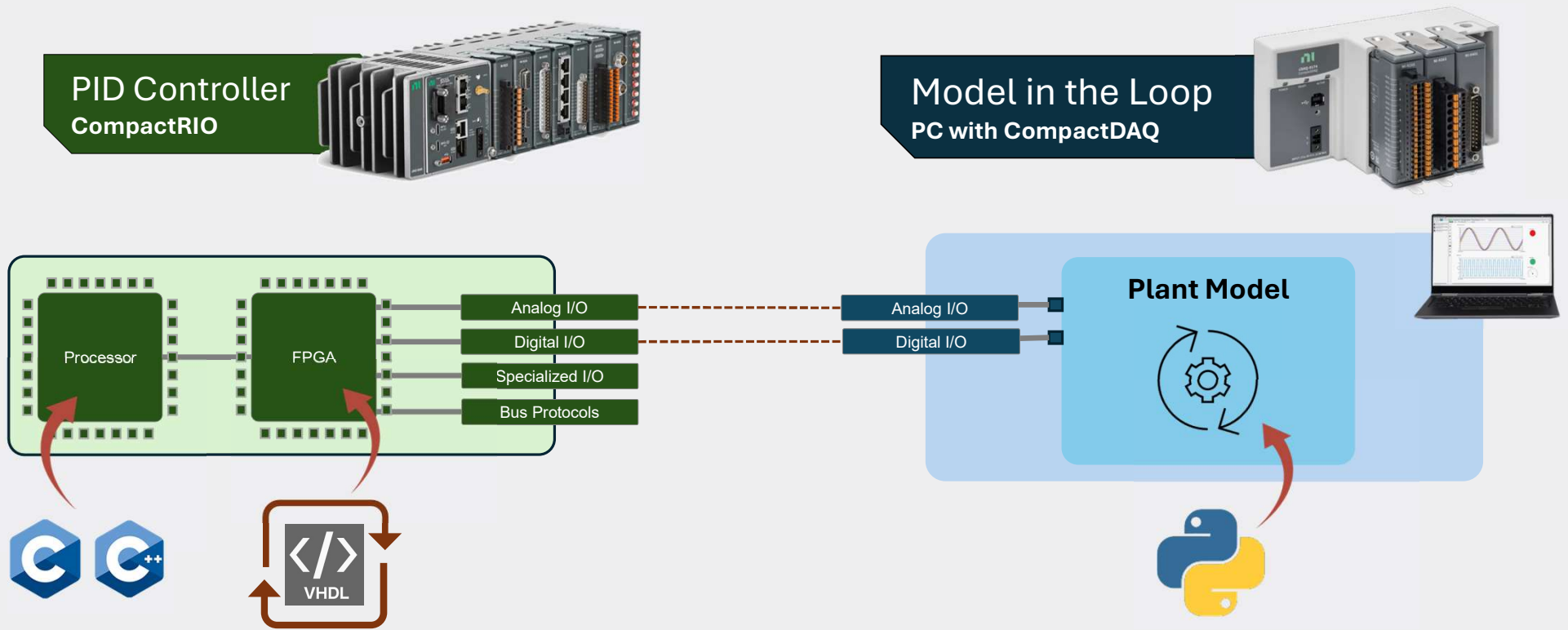


LabVIEW Python Interface

Simplified I/O integration & UI development



System Diagram

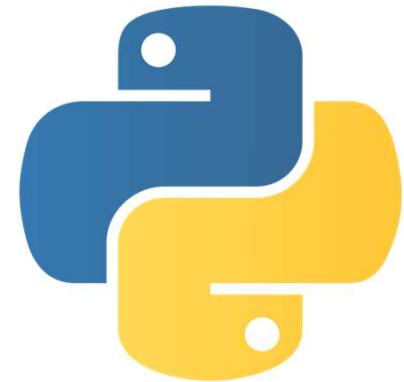


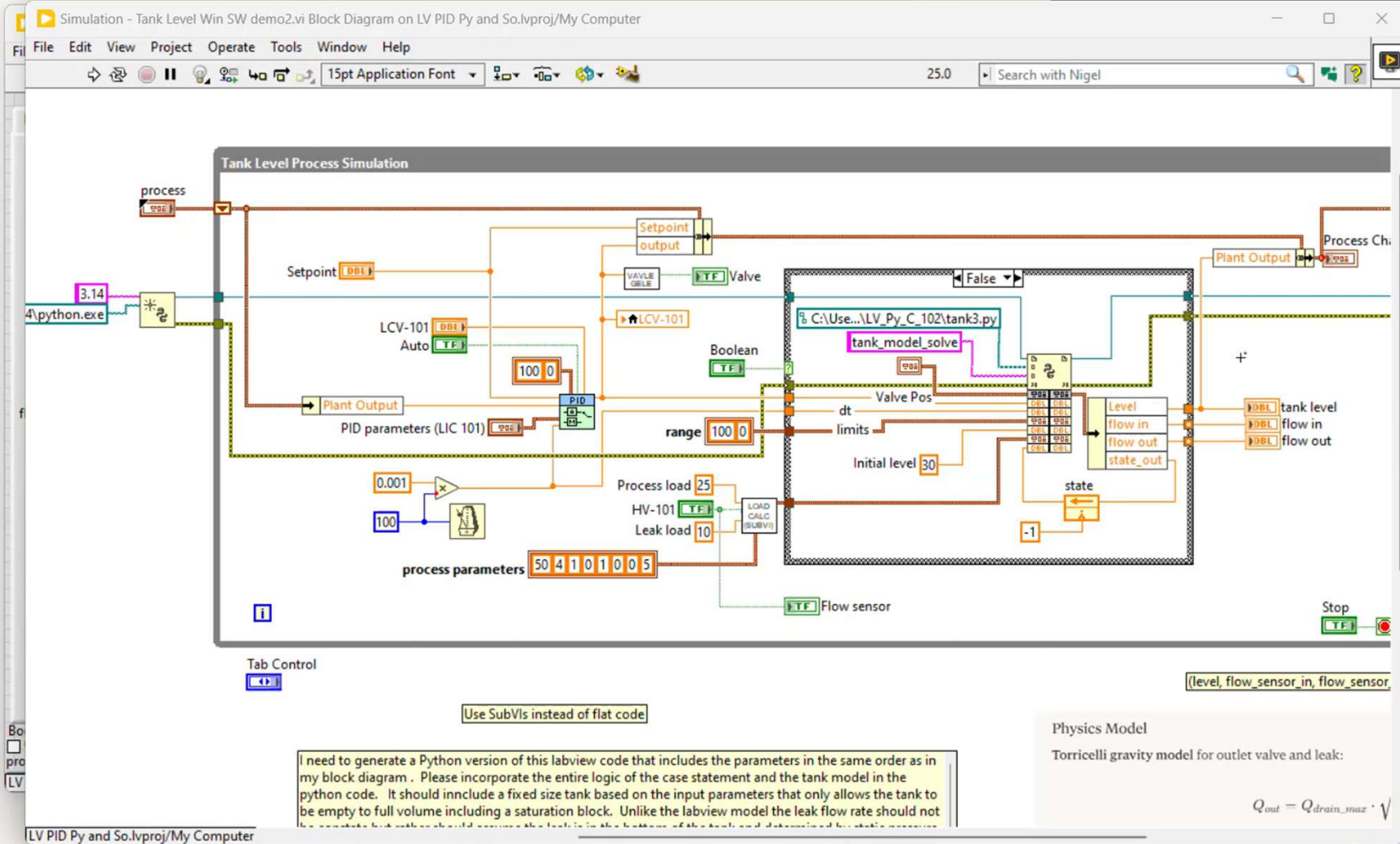
DEMO

```
cl_exec($ch);  
url_getinfo($ch);  
l1['content_type']  
= explode('charset=', $ch['content_type']);  
methd= strtolower($ch['content_type']);  
methd= false;  
= false;  
[]:  
reg_match_all('/(<meta  
charset= strtolower(trim  
($chset!='code-8')  
$new= str_replace  
$html= str_replace  
$chset= $csethd;  
) else  
$chset= false;  
if ($chset=='code-8'  
conv($chset):  
conv($chset):  
conv($chset):
```

Plant Model in Python

Use AI code gen tools to develop a plant model to be incorporated into LabVIEW PID, MIL integration





RIO Platform



Real-Time CPU + FPGA
Versatile, heterogeneous processing

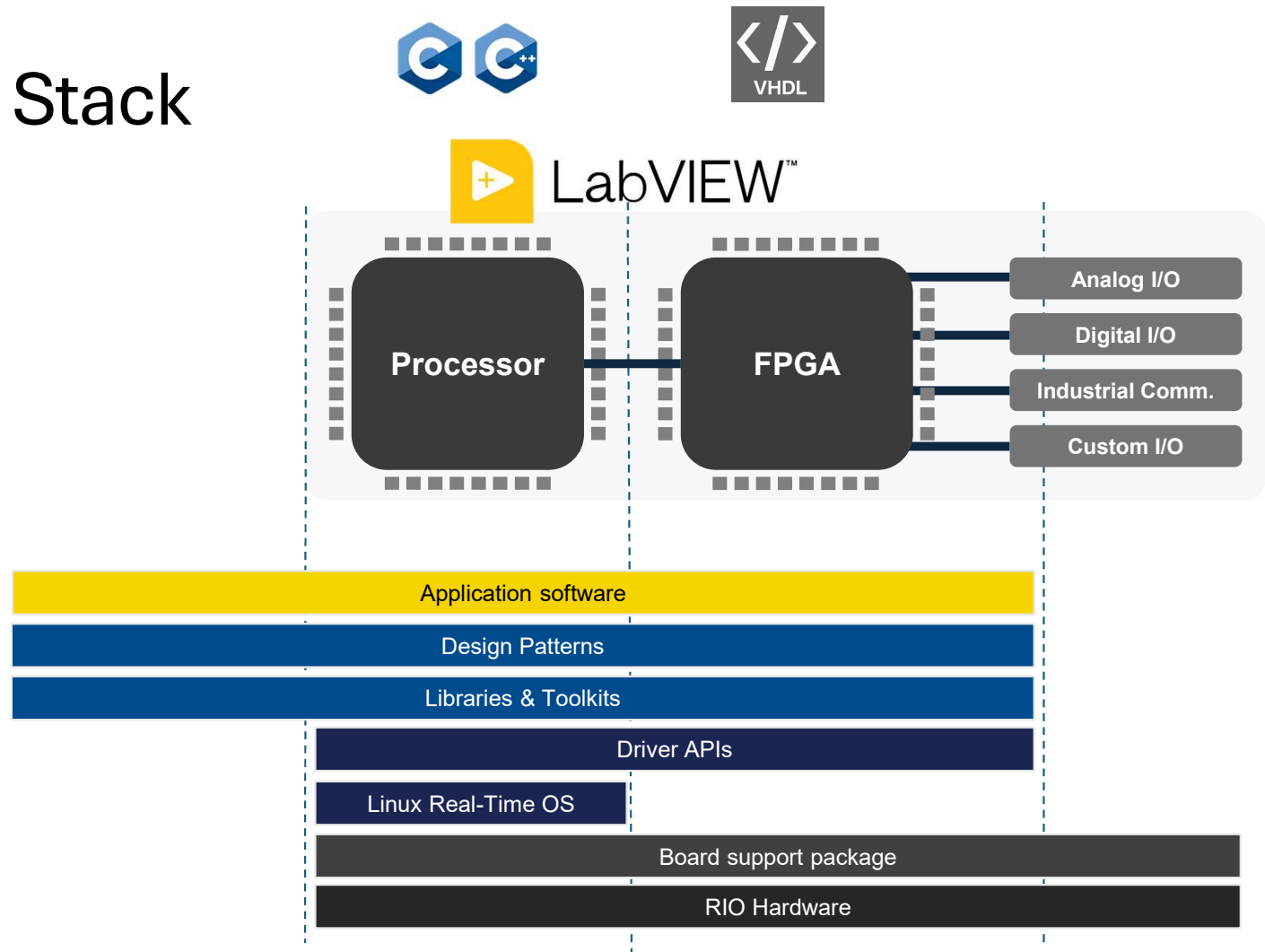
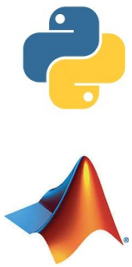
C Series I/O Modules
Measurements, Stimulus, Interfaces



Integrated Development Stack
UI to firmware with open integrations

FPGA in Backplane
Low latency, massively parallel computation

Building the Stack



DEMO

```
private void updateWidget(Context context,
    JodaTimeAndroid.init(context);
    SettingsManager sm = new SettingsManager();
    RemoteViews remoteViews = new RemoteViews();

    // showing dark style if it is
    boolean isDark = sm.loadWidgetID(widgetID);
    if (isDark) remoteViews = new RemoteViews();

    // Register an onClickListener
    Intent main = new Intent(context, MainActivity.class);
    main.setAction(CALL_FROM_WIDGET);
    main.setAction(Long.toString(System.currentTimeMillis()));

    long mId = sm.loadWidgetID(widgetID);

    if (mId > 0) {
        long metricId = setTheValues(remoteViews, mId);
        if (metricId > 0) {
            main.putExtra(Record.class, new Record());
            Log.i(this.getClass().getName(), "Record: " + metricId);
            PendingIntent pendingIntent = PendingIntent.getActivity(context, 0, main, 0);
            remoteViews.setOnClickPendingIntent(widgetID, pendingIntent);
        }
    }
    appWidgetManager.updateAppWidget(widgetID, remoteViews);
}

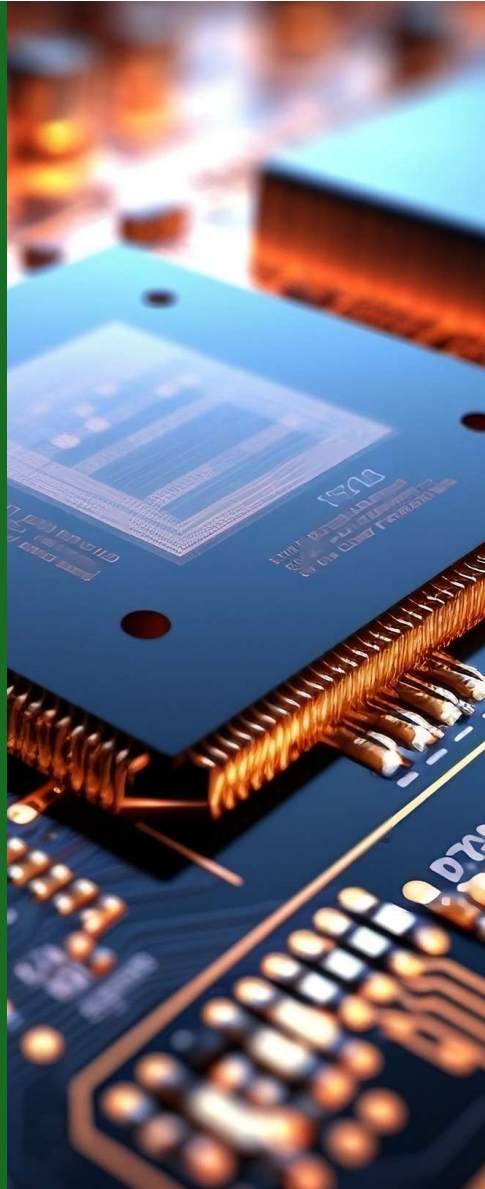
private long setTheValues(RemoteViews remoteViews, long widgetID) {
    DAO datasource = DAO.getInstance(context);
    Record r = datasource.getRecord(widgetID);
    if (r == null) r = datasource.getRecord(widgetID);
    return r.getMetricId();
}
```

Controls Algorithm in C/C++

Use AI code gen tools to develop a controls or simulation algorithm to be incorporated into LabVIEW Real-Time + cRIO application



DEMO



VHDL Generation for FPGA

Use AI code gen tools to develop VHDL code to be compiled and pushed to FPGA target



Test PID.lvproj * - Project Explorer

File Edit View Project Operate Tools Window Help

Items Files

Project Explorer tree

Untitled 47 Front Panel on Test PID.lvproj/FPGA Target

File Edit View Project Operate Tools Window Help

15pt Application Font

Search

Perplexity

Discover Finance Health Academic Patents

perplexity enterprise

Type / for search modes and shortcuts

+ Deep research

Show suggestions

Cyth Systems

Building Demo Summary

Steps

1. Load prompt into AI LLM
2. Respond to question
3. Create IP in Target properties
4. Create CLIP
5. Import Clip
6. Complete code
7. Compile
8. Test

Status			
Complete			
Reports			
Estimated device utilization (synthesis)			
Device Utilization	Used	Total	Percent
Slice Registers	6196	82000	7.6
Slice LUTs	6750	41000	16.5
Block RAMs	5	135	3.7
DSP48s	22	240	9.2



Lessons in VHDL Generation

Generating VHDL without compile or timing errors

Single Cycle Timed Loops are straightforward

5Mhz clock is easier than meeting timing at 40-80 Mhz SCTL rates

XML must have <FormatVersion> tag (the original error)

All constants wire as FXP<32,32> by default - must manually set type

No division inside FPGA CLIP - pre-compute on RT

VHDL must be VHDL-93 compatible for NI Vivado (no declare...begin blocks)

aReset must be boolean data signal, not reset signal type in LV interface

Port signal names: letters/numbers/dashes/underscores only (no parens, spaces, slashes)

Ce clock enable is needed for timed loop integration

FXP type mismatches cause "Error generated from encrypted envelope" in Vivado

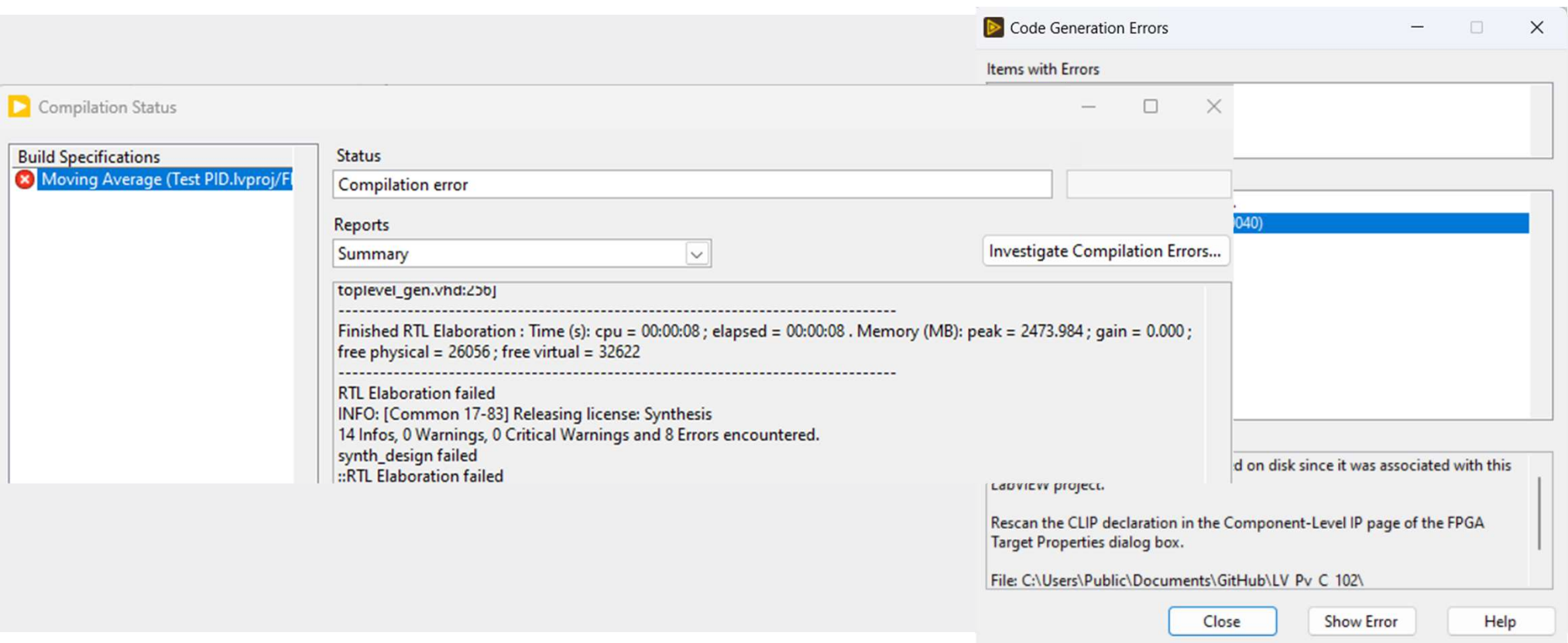
Use std_logic/std_logic_vector only at CLIP boundary, signed internally

Pipeline depth: 3 stages is optimal for 40MHz, 5+ stages for complex operations

Pre-scale Divisions in RT or outside SCTL in LabVIEW

Cocotb + GHDL for pre-synthesis functional validation

Possible but Error prone VHDL generation



The image shows two overlapping windows from the LabVIEW IDE. The 'Compilation Status' window on the left displays a 'Build Specifications' list with 'Moving Average (Test PID.lvproj/F' selected. The 'Status' section shows 'Compilation error'. The 'Reports' section has a 'Summary' dropdown and an 'Investigate Compilation Errors...' button. The main text area shows the output of 'RTL Elaboration failed', including timing and memory statistics, and a list of 8 errors. The 'Code Generation Errors' window on the right is titled 'Items with Errors' and shows a list of errors, with one item partially visible as '(040)'. Below this window, a message box states: 'LabVIEW project. Rescan the CLIP declaration in the Component-Level IP page of the FPGA Target Properties dialog box. File: C:\Users\Public\Documents\GitHub\LV_Pv_C_102\'.

Build Specifications

- Moving Average (Test PID.lvproj/F

Status

Compilation error

Reports

Summary

Investigate Compilation Errors...

RTL Elaboration failed

Finished RTL Elaboration : Time (s): cpu = 00:00:08 ; elapsed = 00:00:08 . Memory (MB): peak = 2473.984 ; gain = 0.000 ; free physical = 26056 ; free virtual = 32622

RTL Elaboration failed

INFO: [Common 17-83] Releasing license: Synthesis

14 Infos, 0 Warnings, 0 Critical Warnings and 8 Errors encountered.

synth_design failed

::RTL Elaboration failed

Code Generation Errors

Items with Errors

(040)

LabVIEW project.

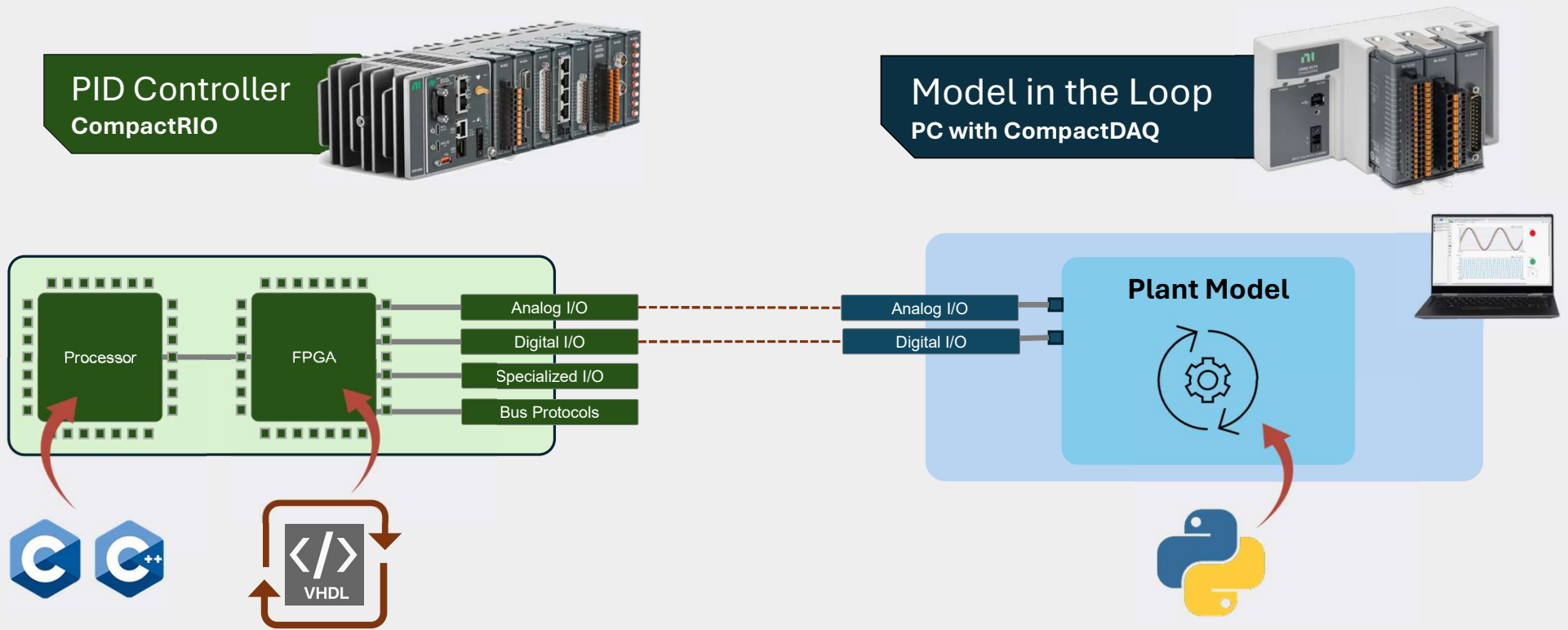
Rescan the CLIP declaration in the Component-Level IP page of the FPGA Target Properties dialog box.

File: C:\Users\Public\Documents\GitHub\LV_Pv_C_102\

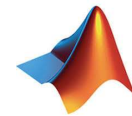
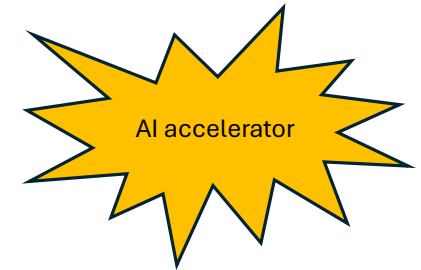
Close Show Error Help

LabVIEW Upgrade Breaks CLIP

System Diagram



Right Tool, Right Job



	VHDL	Python	C / C++ / C#	Simulink	LabVIEW
FPGA HDL 	✓	-	-	✓	✓
Controller I/O integration	✓	-	-	-	✓
Build & deploy to target	✓		✓		✓
Algorithm engineering 	-	✓	✓	✓	✓
Test bench I/O integration	-	✓	✓	-	✓
Data processing 	-	✓	✓	✓	✓
File I/O and databasing 	-	✓	✓	✓	✓



Validation

Simulation, test bench setup,
benchmarking



Validation & Benchmarking

Generating unit test vectors via AI prompt

Performance benchmarking → test bench in LabVIEW

- VI Analyzer
- Timing & Analyzer
- Nigel code analysis

Source code control

Breaking down the workflow

